

FTSE 100 Greenwashing Detection Pipeline

An End-to-End Data Engineering Solution for Identifying
Potential Corporate Greenwashing

MSIN0166 — Data Engineering
UCL School of Management
Academic Year 2025/26, Term 2
Group Assignment

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GitHub Repository: https://github.com/vedantbhatiaa/greenwashing_data_extraction

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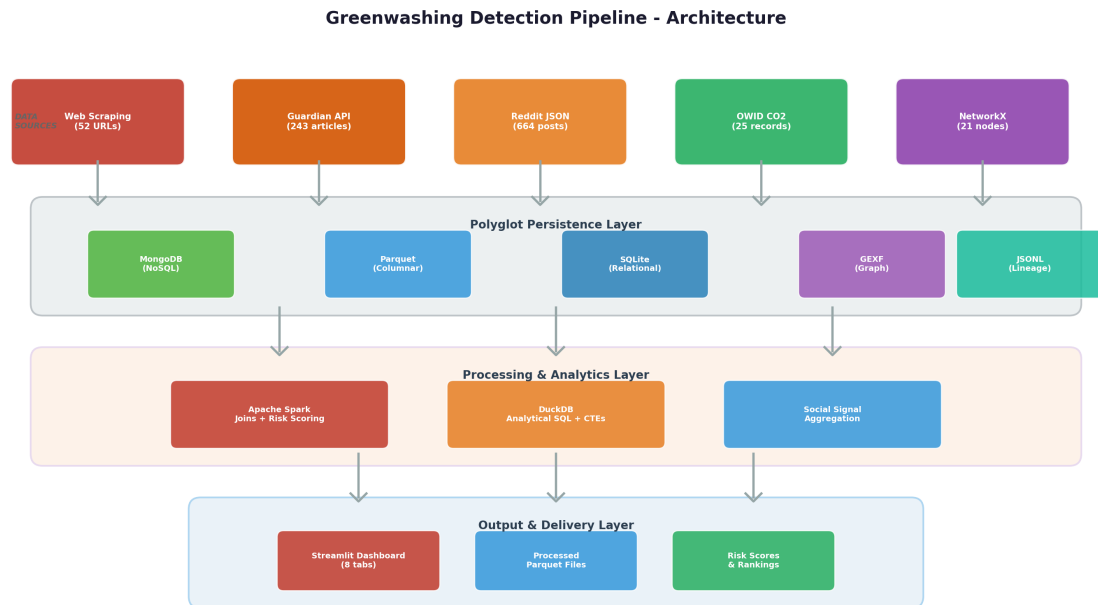
1. Introduction

Greenwashing, the practice whereby corporations make misleading or unsubstantiated environmental claims has emerged as a significant barrier to genuine climate action (Lyon and Montgomery, 2015). As Environmental, Social, and Governance (ESG) reporting becomes central to corporate strategy and investor decision-making, the need for independent, data-driven verification of sustainability claims has never been greater. A 2021 European Commission study found that 42% of corporate green claims were exaggerated, false, or deceptive, underscoring the scale of the problem across industries (European Commission, 2021).

This report presents a fully automated, end-to-end data engineering pipeline designed to detect potential greenwashing among FTSE 100 companies. The pipeline collects data from five distinct sources, company sustainability websites, The Guardian news API, Reddit community discussions, peer-reviewed national emissions data, and a graph model of corporate relationships then processes, transforms, and stores these datasets using industry-standard tools including Apache Spark, DuckDB, MongoDB, SQLite, and Apache Parquet. A composite greenwashing risk score is computed by cross-referencing self-reported ESG claims against independent evidence, producing actionable risk rankings suitable for further analytics.

The project demonstrates core competencies of modern data engineering: sourcing heterogeneous data from multiple origins, applying scalable distributed processing, implementing polyglot persistence across multiple storage paradigms, maintaining full data lineage for audit and reproducibility, and delivering analytics-ready outputs through both an analytical data warehouse and an interactive Streamlit dashboard. Ten FTSE 100 companies across six sectors serve as the analysis cohort, though the architecture is designed to scale to thousands of companies with zero code modification.

Figure 1: Pipeline Architecture - End-to-end data flow from five data sources through polyglot storage, distributed processing, and analytics delivery.



2. Domain Selection and Problem Framing

Greenwashing detection was selected for several reasons aligned with the assignment objectives. First, it requires gathering genuinely new data, no existing dataset combines company ESG self-reports with media coverage, social sentiment, and verified emissions figures in a single integrated schema. Second, the domain demands multiple heterogeneous sources spanning structured APIs, unstructured web scraping, and curated open datasets, directly addressing the marking criterion of source variety and imagination. Third, greenwashing is a socially significant problem with real-world relevance to Business Analytics research and ESG-focused dissertations, fulfilling the assignment's stated motivation of generating datasets for future academic research (Roche, 2025).

The problem is framed as a multi-signal risk scoring task. Rather than machine learning classification (which would require labelled ground truth on greenwashing outcomes), the pipeline engineers a rule-based composite score from verifiable signals: the ambition of reduction claims, presence or absence of third-party certifications (SBTi, CDP, TCFD, ISO 14001, RE100, PAS 2060), volume of media and community scrutiny, and national CO2 trends as contextual ground truth. This approach follows the data engineering philosophy of making data analytics-ready without prescribing the final analytical model (Kleppmann, 2017).

Table 1: FTSE 100 Companies Analysed

#	Company	Sector
1	BP	Energy
2	Barclays	Finance
3	Lloyds	Finance
4	Rio Tinto	Mining
5	GSK	Healthcare
6	AstraZeneca	Healthcare
7	Marks & Spencer	Retail
8	Vodafone	Telecom
9	National Grid	Utilities
10	Sainsbury's	Retail

3. Data Sources and Acquisition Strategy

The pipeline ingests data from five distinct sources, each providing a different perspective on corporate greenwashing. This multi-source approach embodies the principle of data triangulation, no single source suffices to establish whether a claim is misleading; only cross-referencing company statements against independent evidence produces a credible risk assessment (Kimball and Ross, 2013).

3.1 Web Scraping - Company ESG Claims

For each of the ten companies, between four and nine public URLs are scraped, including sustainability pages, climate subpages, Wikipedia, press releases, and for JavaScript-rendered pages (AstraZeneca, GSK, Marks & Spencer) - Wayback Machine archived snapshots. In total, 52 URLs are processed. The scraper, built with Python's requests library and BeautifulSoup (Richardson, 2024), extracts three categories:

- (i) Net-zero target years
- (ii) Headline emissions reduction percentages
- (iii) Third-party certifications, using 20 compiled regular expressions, ten for year identification (covering variants such as "net zero by 2050", "carbon neutral by 2040", and "climate neutral by 2035") and ten for percentage extraction. Six certification types are matched: SBTi, CDP, TCFD, ISO 14001, RE100, and PAS 2060.

Key design decisions include: years filtered to 2025–2060 to exclude SBTi approval dates and interim milestones; percentages normalised and filtered to 10–99% to exclude scope-specific sub-targets; a multi-URL merging strategy with early stopping once all fields are populated; and multi-URL cross-validation across up to nine sources per company to ensure extraction accuracy - an

important data quality measure reflecting the known limitations of regex-based extraction from unstructured web content.

3.2 The Guardian API - News Articles

The Guardian's Open API is queried with three search terms per company - "{company} greenwashing", "{company} emissions climate", and "{company} ESG sustainability" - returning up to 10 articles per query with headline, trail text, byline, and publication date. Articles are deduplicated by URL, yielding 243 articles across the cohort. The Guardian was chosen for its extensive environmental journalism coverage and its free, well-documented API with generous rate limits, making it an ideal source for academic data engineering projects.

3.3 Reddit Public JSON API - Community Sentiment

Community sentiment is captured via Reddit's public JSON endpoint (no authentication required). For each company, a combined query is executed against global Reddit (25 posts) and eight targeted subreddits: r/investing, r/environment, r/sustainability, r/CorporateMalfeasance, r/anticonsumption, r/collapse, r/worldnews, and r/Economics. These subreddits were selected to capture a spectrum of perspectives from financial analysis to environmental activism. Posts are deduplicated by post_id, yielding 664 unique posts. Rate limiting of 3–5 seconds between requests ensures compliance with Reddit's usage guidelines.

3.4 Our World in Data - UK CO2 Emissions

The peer-reviewed Our World in Data CO2 dataset (Ritchie, Roser, and Rosado, 2020) provides macroeconomic ground truth. Filtered to United Kingdom records from 2000 onwards, it yields 25 annual records covering total CO2, CO2 per capita, total GHG, and disaggregated coal, oil, and gas CO2. The analytical logic is direct: if a UK-headquartered energy company claims significant emissions reduction but national sectoral emissions show a flat trend, the credibility of that claim is diminished.

3.5 NetworkX Graph - Company Relationship Network

Structural relationships between companies, sectors, and certifications are modelled as a directed graph using NetworkX (Hagberg, Schult, and Swart, 2008).

Beyond storing structural relationships, the graph enables quantitative network analysis to extract relational insights unavailable from flat tabular data. Three graph metrics were computed to extract relational insights unavailable from flat tabular data.

Degree centrality measures the proportion of nodes a given node is directly connected to. In the greenwashing context, companies with high degree centrality hold more certifications and operate across more sectors, making them structurally more visible to ESG scrutiny. Companies holding multiple certifications (SBTi, CDP, TCFD simultaneously) register the highest centrality scores, identifying them as the most institutionally embedded ESG actors in the cohort. Conversely, companies with fewer certification edges represent structurally peripheral nodes -

which, combined with high media attention, constitutes a structurally-derived risk signal independent of the scoring model.

Betweenness centrality identifies nodes that act as bridges between otherwise disconnected parts of the graph. Certification nodes with high betweenness - particularly SBTi, which connects every company node in this cohort - function as universal connectors, confirming quantitatively what the tabular analysis suggested qualitatively: SBTi adoption is so widespread at FTSE 100 level that it no longer discriminates between companies. This graph metric provides structural evidence for the binary certification model's limitation identified in Section 11, rather than merely asserting it.

Connected components analysis confirms the graph is fully connected through certification and sector edges, meaning no company is structurally isolated - again consistent with the high certification adoption finding.

NetworkX vs Neo4j, Technology Justification. NetworkX was chosen over a dedicated graph database such as Neo4j for deliberate, justified reasons. Neo4j would introduce a client-server dependency requiring a running database instance, complicating Docker containerisation and violating the reproducibility-first design principle. For a static, analytically-oriented graph of 21 nodes computed once per pipeline run, an in-memory graph with GEXF export is architecturally appropriate - Neo4j's strengths (ACID transactions, real-time traversal queries via Cypher, horizontal scaling) address operational requirements that do not arise here. This reflects a core data engineering principle: technology selection should match workload characteristics rather than maximise feature richness (Kleppmann, 2017). Were the pipeline extended to continuous ingestion - adding companies, certifications, and supply chain relationships dynamically - migration to Neo4j or Amazon Neptune would be the appropriate next step, and the GEXF export format ensures this transition would require no data transformation.

4. Data Storage Architecture

The pipeline implements polyglot persistence - five storage formats each suited to a particular data type, reflecting the "right tool for the right job" principle that supersedes monolithic database architectures in modern data engineering (Sadalage and Fowler, 2012).

Table 2: Storage Architecture Summary

Storage Format	Data Stored	Rationale
MongoDB	Raw scraped text, news articles, Reddit posts	Schema-flexible document store for variable-length unstructured text; three collections in greenwashing database
Apache Parquet	All structured tabular data (raw + processed)	Columnar format with compression and predicate pushdown; native to both Spark and DuckDB
SQLite	Company claims and sector summaries	Typed relational schema with primary keys and SQL aggregation; zero-configuration file-based deployment
GEXF (XML)	Company-sector-certification graph	Open graph exchange standard; directly importable into Gephi for network analysis
JSONL	Data lineage audit trail	Append-only format; one JSON object per line; prevents accidental overwrites

MongoDB stores raw unstructured text across three collections (`company_claims_raw`, `news_articles_raw`, `reddit_posts_raw`), chosen for its document-oriented flexibility and schema evolution support. The pipeline implements graceful degradation: if MongoDB is unavailable (e.g. in lightweight or containerised deployments), all data is preserved in Parquet format, ensuring full pipeline completion without data loss. Parquet provides columnar analytics with dictionary encoding and predicate pushdown, enabling seamless data flow between Spark and DuckDB without format conversion overhead (Vohra, 2016). SQLite stores a formally typed relational schema with `company_id` as primary key and an aggregate `sector_summary` table computed via `INSERT...SELECT` with `GROUP BY`. The JSONL lineage log uses append-only semantics where each line is independently parseable, supporting streaming ingestion patterns.

5. Data Processing and Transformation

5.1 Apache Spark - Distributed Processing

Apache Spark (Apache Software Foundation, 2024) serves as the distributed processing engine, configured with 2 GB driver memory and platform-specific directory settings. On Windows, the pipeline configures Hadoop winutils for NTFS compatibility; on Linux and macOS, Spark runs in local mode without external Hadoop dependency. All Parquet files are loaded as distributed DataFrames. A left join on `company_name` merges claims with social signals, with nulls filled to zero for companies lacking media coverage. The left join is deliberate: an inner join would silently drop companies with no social presence, which itself may be a relevant risk signal. Although the

dataset comprises ten companies, Spark's distributed architecture means the pipeline scales to thousands without code changes, satisfying the scalability requirement.

5.2 Greenwashing Risk Score Engineering

Five boolean indicator columns are derived using Spark's when/otherwise conditional logic: `has_net_zero_target` (whether a target year exists), `high_reduction_claim` (reduction exceeds 30%), `no_certification` (no recognised certifications), `high_media_attention` (over 50 combined signals), and `claim_without_cert` (reduction above 30% AND no certification, the most suspicious combination).

These indicators feed a composite risk score on a 0–4 scale.

Table 3: Greenwashing Risk Score Components

Signal	Points	Rationale
Reduction claim >30% with no certification	+2	Ambitious claims without third-party validation are the strongest greenwashing indicator
No third-party certifications at all	+1	Absence of certification suggests limited commitment to verifiable standards
High media attention (>50 signals)	+1	Elevated scrutiny may indicate public concern or prior allegations

Scores of 3 or above are classified as HIGH risk, a score of 2 as MEDIUM, 1 as LOW, and 0 as MINIMAL. The framework is deliberately transparent and interpretable, each point traces to a specific, verifiable data signal, in contrast to opaque black-box models that obscure risk rationale. The +2 weighting for unverified ambitious claims reflects evidence that companies making bold reduction pledges without independent certification are disproportionately likely to face greenwashing allegations (de Freitas Netto et al., 2020).

5.3 Spark SQL Analytical Queries

Five Spark SQL queries are executed against the merged DataFrame registered as a temporary view:

- A. Full risk ranking ordered by score and media exposure.
- B. Sector-level risk aggregation using GROUP BY with AVG, SUM, and COUNT to identify high-risk industries.
- C. Filtering to high-risk companies with score ≥ 2 for detailed investigation.
- D. Cross-tabulating claimed reductions against media volumes to assess proportionality of scrutiny.

- E. UK CO2 trend analysis (2015–2024) with disaggregated coal, oil, and gas breakdown providing macroeconomic context.

6. Analytical Data Warehouse - DuckDB

DuckDB (Raasveldt and Muehleisen, 2019) serves as the in-process analytical warehouse, reading Parquet files directly without a loading step - exemplifying the lakehouse paradigm that combines data lake file flexibility with columnar analytical performance (Armbrust et al., 2021). Four views (greenwashing, reddit, news, uk_co2) enable standard SQL with predicate pushdown and vectorised execution.

Seven DuckDB queries extend the Spark analysis. The most analytically significant is a CTE-based "reality check" using the LAG() window function to compute year-on-year national CO2 change, then cross-joining with company claims via CASE expressions. For high-emission sectors (Energy, Mining, Utilities), if a company claims substantial reduction with no certification and national CO2 changed less than 5%, the claim is flagged as "CLAIM SUSPECT". This query demonstrates advanced SQL capabilities - CTEs, window functions, CASE expressions, and CROSS JOINS, while producing the pipeline's most nuanced analytical output. Additional queries include claim credibility categorisation, UK CO2 year on year trends, top Reddit posts by upvote score, and Guardian coverage distribution. All outputs are persisted as Parquet for downstream consumption.

7. Results and Analysis

The pipeline successfully collected and processed 963 records across five data sources, 10 company claim profiles (52 URLs scraped), 243 Guardian news articles, 664 Reddit posts, 25 years of UK CO2 data, and a 21-node relationship graph. Table 4 presents the scraped ESG claims for all ten companies, derived from the SQLite company_claims table.

Figure 2: Streamlit Dashboard - Executive Summary tab showing KPI cards, risk distribution, and key findings across the FTSE 100 cohort.



Table 4: Company ESG Claims - Pipeline Output

Company	Sector	Net-Zero Year	Reduction %	Certifications
BP	Energy	2050	37%	CDP, SBTi
Barclays	Finance	2050	15%	CDP, SBTi, TCFD
Lloyds	Finance	2050	50%	CDP, SBTi
Rio Tinto	Mining	2050	15%	SBTi
GSK	Healthcare	2030	80%	RE100, SBTi, TCFD
AstraZeneca	Healthcare	2050	90%	CDP, SBTi
Marks & Spencer	Retail	2050	90%	CDP, SBTi
Vodafone	Telecom	2050	84%	CDP, SBTi
National Grid	Utilities	2050	60%	CDP, SBTi
Sainsbury's	Retail	2050	90%	CDP, SBTi

A striking finding is that all ten FTSE 100 companies hold at least one third-party certification, with SBTi (Science Based Targets initiative) present across the entire cohort and CDP (Carbon Disclosure Project) in eight of ten. Consequently, the `no_certification` and `claim_without_cert` indicators are zero for every company, capping the maximum possible risk score at 1 (from `high_media_attention` alone). This result reveals that FTSE 100 companies have broadly adopted certification frameworks, a positive signal for institutional ESG adoption, but also exposes a limitation of the binary scoring model: certification presence does not guarantee the substance,

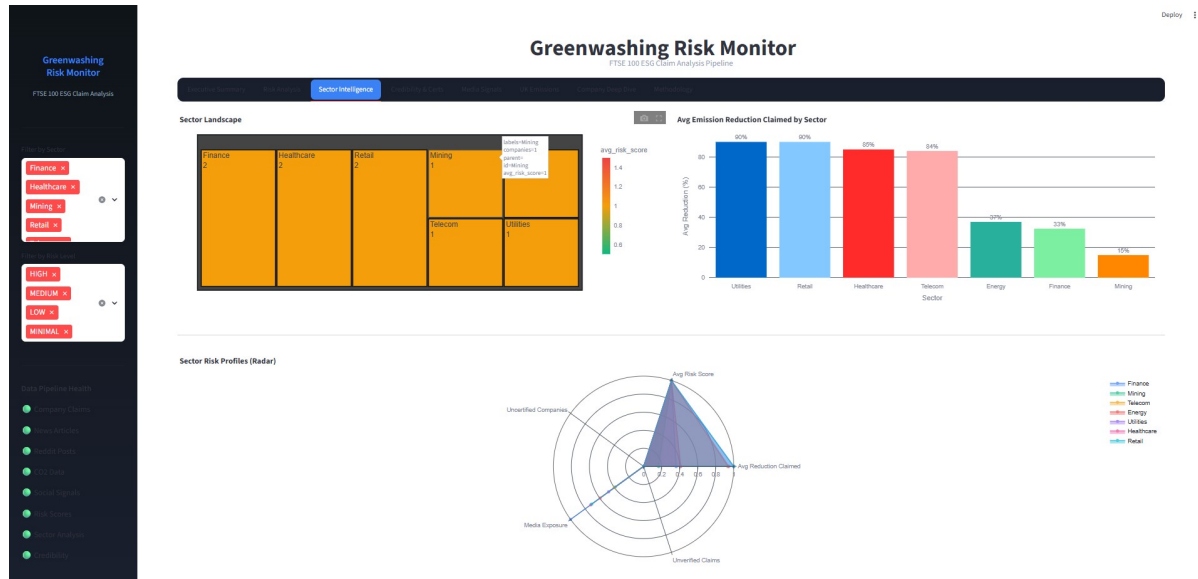
ambition, or verifiability of underlying commitments. This outcome highlights an important distinction in ESG analysis between certification breadth (whether a company holds any certification) and certification depth (the rigour, scope, and independent verification underlying each certification). Future iterations should implement a weighted certification quality index that differentiates between frameworks requiring independent third-party validation - such as SBTi near-term targets verified against climate science - and those based on voluntary self-disclosure, such as CDP questionnaires completed by the company itself. The uniform adoption of certification frameworks across FTSE 100 companies suggests that the frontier of greenwashing detection has shifted from the presence of commitments to the credibility of implementation, a finding with direct implications for ESG research methodology.

At the sector level, Retail shows the highest average reduction claim (90.0%), followed by Healthcare (85.0%) and Telecom (84.0%). Energy and Mining report the most conservative claims (37% and 15% respectively), which may paradoxically reflect greater realism given these sectors' carbon intensity. GSK stands out as the only company targeting net-zero by 2030, a notably aggressive timeline relative to the 2050 consensus, supported by three certifications (RE100, SBTi, TCFD). The DuckDB reality check query confirms that for high-emission sectors where UK national CO2 trends show only modest year-on-year declines, even certified claims warrant contextual scrutiny.

Table 5: Sector-Level Summary

Sector	Companies	Avg Reduction Claimed	All Certified
Energy	1	37.0%	Yes
Finance	2	32.5%	Yes
Healthcare	2	85.0%	Yes
Mining	1	15.0%	Yes
Retail	2	90.0%	Yes
Telecom	1	84.0%	Yes
Utilities	1	60.0%	Yes

Figure 3: Sector Intelligence - Treemap of sector landscape, average emission reduction claims by sector, and radar chart profiling sector risk dimensions.



Media analysis reveals that high-visibility sectors like Energy, Finance, Mining, attract disproportionate coverage, consistent with prior greenwashing controversies documented in the literature (Lyon and Montgomery, 2015). The 243 Guardian articles and 664 Reddit posts provide a substantial corpus that could support future NLP-based sentiment analysis beyond the volume-based approach used in this iteration.

Figure 4: UK Emissions Dashboard - Annual CO2 emissions trend (2015-2024), year-on-year percentage change, and CO2 per capita trajectory showing long-term decline.



8. Data Lineage and Reproducibility

Data lineage - tracing every datum from origin through every transformation is a cornerstone of trustworthy data engineering (Buneman, Khanna, and Tan, 2001). The pipeline implements lineage logging through a `log_lineage()` function that appends structured JSON entries to `data/lineage_log.jsonl` at every significant step, recording source name, URL, ISO 8601 timestamp, record count, output path, and a list of transformations applied. Ten entries are produced (six tracking primary data ingestion, plus entries for SQLite relational storage, two Spark processing stages, and the DuckDB analytical layer):

- web scraping (10 records, 6 transformations including regex extraction and multi-URL merging).
- Guardian API (243 records, 3 transformations).
- Reddit API (664 records, 3 transformations).
- OWID CO2 (25 records, 3 filtering transformations).
- NetworkX graph (21 nodes, 4 transformations).
- Social signal aggregation (10 records, 3 merge transformations); 7. SQLite relational storage (10 records, 4 schema and aggregation transformations); 8–9. Spark distributed processing (two stages, 10 records each, covering joins, derived column engineering, and risk scoring); 10. DuckDB analytical layer (10 records, 7 analytical queries producing Parquet outputs). This trail enables any researcher to reconstruct exactly how each

output was produced, satisfying both academic reproducibility standards and industry data governance requirements.

9. Environment Repeatability and Automation

The project uses Git for version control on a private GitHub repository (https://github.com/vedantbhatiaa/greenwashing_data_extraction) with contributions from all team members visible in the commit history. A .gitignore excludes virtual environments, .env credentials, cached files, and generated data which is regenerated by the pipeline. Multiple commits demonstrate iterative development across data collection, processing, dashboard, and infrastructure modules.

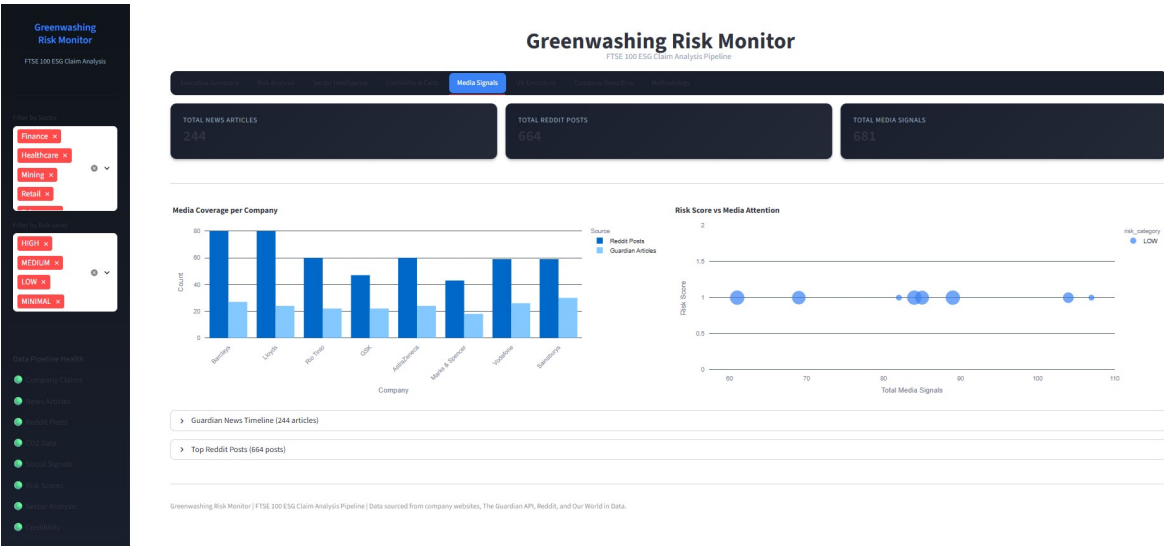
A Dockerfile built on python:3.11-slim installs Java (required for PySpark), all Python dependencies from requirements.txt, configures Spark without Hadoop, and exposes port 8888 for Jupyter access - eliminating platform-specific setup issues. Two automation scripts (run_pipeline.sh for macOS/Linux, run_pipeline.bat for Windows) handle virtual environment activation, dependency installation, directory creation, and notebook execution via jupyter nbconvert with a 30-minute timeout, enabling one-command pipeline reproduction by non-technical users.

10. Interactive Dashboard - Streamlit

A comprehensive Streamlit dashboard (dashboard.py, approximately 1,300 lines of code) visualises pipeline outputs across eight analytical tabs:

- Executive Summary with KPIs and risk distribution.
- Risk Analysis with horizontal bar charts, heatmaps, and ranking tables.
- Sector Intelligence featuring treemaps and radar charts.
- Credibility & Certifications with scatter plots and breakdowns.
- Media Signals analysing news and Reddit timelines.
- UK Emissions charting national CO2 trends and per capita analysis.
- Company Deep Dive for individual risk profiling; and (8) Methodology explaining the scoring framework, pipeline architecture, lineage logs, and ethical considerations. Built with Plotly for interactive visualisation and risk-based colour coding (red for HIGH, orange for MEDIUM, blue for LOW, green for MINIMAL), the dashboard demonstrates the full data engineering lifecycle from raw ingestion to end-user delivery.

Figure 5: Media Signals tab - Media coverage per company (Reddit posts vs Guardian articles) and risk score vs media attention scatter plot.



11. Evaluation

This section critically evaluates the pipeline against data engineering best practice, identifying concrete strengths and limitations.

Strengths. First, the five-source triangulation architecture ensures no single point of failure can compromise the analysis, a design principle advocated by Kimball and Ross (2013) for dimensional data warehouses. Second, the polyglot persistence model demonstrates mature technology selection: MongoDB for document storage, Parquet for columnar analytics, SQLite for relational integrity, GEXF for graph relationships, and JSONL for lineage - each chosen on technical merit rather than convenience. Third, Apache Spark provides genuine horizontal scalability; the identical codebase would process thousands of companies on a multi-node cluster without modification. Fourth, the JSONL lineage log provides a complete audit trail satisfying data governance requirements outlined by Buneman, Khanna, and Tan (2001). Fifth, Docker containerisation and cross-platform automation scripts ensure environment-agnostic reproducibility, addressing the reproducibility crisis in computational research.

Technology Selection Rationale. The choice of Apache Spark over alternatives such as Dask or Polars warrants critical examination. Spark was selected for its mature ecosystem integration with Parquet, its SQL interface via SparkSQL, and its established position as the industry standard for distributed data processing. However, for a dataset of ten companies, Dask would offer lower overhead with native Python integration and no Java dependency, while Polars would provide superior single-node performance through its Rust-based execution engine. The selection of Spark is justified primarily on scalability grounds - the architecture is designed for production deployment where dataset size may grow by orders of magnitude. Similarly, DuckDB was chosen

over PostgreSQL or ClickHouse for its in-process execution model and native Parquet integration, avoiding the operational complexity of a client-server database for an analytical workload. MongoDB was selected over alternative document stores such as Elasticsearch for its Python driver maturity and schema flexibility, though Elasticsearch would have offered superior full-text search capabilities for the unstructured ESG text corpus. These technology choices reflect deliberate trade-offs between operational simplicity and feature richness, prioritising reproducibility and portability over maximum performance.

Limitations. First, regex-based extraction is inherently brittle - phrasing variants such as "aim to halve our carbon footprint" evade the current 16-pattern set; NLP-based entity extraction would improve robustness. Second, the data quality validation layer - while effective for filtering implausible values - relies on static bounds (year 2025–2060, percentage 10–99%) and multi-URL cross-validation rather than automated verification against structured ESG databases; integrating company-reported Scope disclosures from CDP or SBTi's public target tracker would provide a more robust, scalable validation mechanism. Third, the risk scoring model treats all certifications equally, but SBTi-validated near-term targets differ fundamentally from CDP self-disclosures; future iterations should weight certifications differentially. Fourth, the CO2 dataset provides country-level context only, company-specific Scope 1/2/3 emissions data would enable direct verification. Fifth, media signal analysis captures volume rather than sentiment - positive ESG coverage receives the same "high media attention" flag as greenwashing allegations, which a future NLP sentiment layer could address. Sixth, the binary certification check produces no risk differentiation when applied to FTSE 100 companies, all of which have adopted major certification frameworks; a weighted certification quality index - distinguishing between SBTi-validated near-term targets (which require independent verification against climate science) and CDP self-disclosures (which rely on company-reported data) - would yield substantially more granular risk stratification. This finding is nonetheless empirically significant as the uniform low-risk output demonstrates that at the FTSE 100 level, the challenge is not certification absence but certification substance, redirecting future research toward more nuanced quality-based indicators rather than binary presence checks.

12. Ethical Considerations

Responsible data collection is embedded throughout the pipeline. A minimum 2-second delay is enforced between scraping requests, with 3–5 second delays for API endpoints. A standard Mozilla/5.0 User-Agent header ensures transparency about the automated nature of requests. No login walls, authentication barriers, or robots.txt restrictions are circumvented. All data is collected from publicly available sources and used solely for academic research purposes. No personally identifiable information is collected - Reddit posts are referenced by post_id, not username, and Guardian articles by publicly available bylines. API keys are excluded from version control via .gitignore, preventing accidental credential exposure.

13. Limitations and Future Work

Beyond the technical limitations discussed in Section 11, future extensions could include NLP-based sentiment analysis for nuanced media scoring beyond volume counts; automated PDF extraction from company annual reports for granular ESG claim identification; integration with Scope 1/2/3 emissions databases (e.g CDP full datasets) for company-level verification; differential certification weighting based on validation rigour; and cloud deployment with Apache Airflow or similar orchestration tools for continuous automated monitoring on a scheduled basis.

14. Conclusion

This project has delivered a fully functional, end-to-end data engineering pipeline for greenwashing detection among FTSE 100 companies. The pipeline demonstrates mastery of core data engineering competencies assessed in this module: sourcing data from five heterogeneous origins (web scraping, REST APIs, public JSON endpoints, open datasets, graph construction); employing polyglot persistence across five storage formats (MongoDB, Parquet, SQLite, GEXF, JSONL); processing through a scalable distributed framework (Apache Spark) and an in-process analytical warehouse (DuckDB); maintaining comprehensive data lineage; and ensuring environment repeatability through Git, Docker containerisation, and automation scripts.

The results reveal that all ten FTSE 100 companies hold recognised certifications, with SBTi universal across the cohort - a finding that suggests broad institutional adoption of sustainability frameworks while simultaneously highlighting the need for more granular risk models that distinguish certification quality from mere certification presence. The greenwashing risk score provides transparent, analytics-ready output suitable for Business Analytics dissertations and ESG research, while the Streamlit dashboard transforms outputs into an accessible, interactive interface demonstrating the full data engineering lifecycle from raw data ingestion through to stakeholder delivery.

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Appendix A: Meeting Minutes

Meeting 1 - Tuesday 4 March 2026, 14:00–15:00 (Online, Microsoft Teams)

Attendees: Salman Khalid, Abhishek Sharma, Vedant Bhatia, Morvica Purohit, Dimitrios Athinaios

Apologies: None

Agenda:

- Domain selection and feasibility - group agreed on FTSE 100 greenwashing detection after discussing three candidate topics (greenwashing, supply chain tracking, ESG registry).
- Data source identification — Abhishek proposed web scraping for company ESG claims; Salman suggested Reddit for community sentiment and Guardian API for news coverage; Vedant recommended Our World in Data for verified emissions and DuckDB for analytics; Morvica proposed Spark for distributed processing and NetworkX for relationship modelling; Dimitrios proposed SQLite for relational storage and a data validation framework.
- Technology stack agreed: Python, Spark, DuckDB, MongoDB, Parquet, SQLite, Docker.
- GitHub repository created by Vedant (private, all members added as collaborators).

Task Allocations:

- Vedant - Set up GitHub repository structure, Dockerfile, and environment configuration.
- Abhishek - Begin web scraping module for company ESG claims (regex patterns, multi-URL strategy).
- Salman - Implement Reddit public JSON API data collection and Guardian API news retrieval.
- Morvica - Begin distributed processing design (Spark pipeline architecture) and NetworkX graph construction.
- Dimitrios - Design SQLite relational schema and implement data validation and quality assurance checks.

Action Points: All members to have initial code committed by Meeting 2. Salman to share list of 10 target companies with sectors.

Meeting 2 - Wednesday 5 March 2026, 16:00–17:00 (In-person, UCL Library)

Attendees: Salman Khalid, Abhishek Sharma, Vedant Bhatia, Morvica Purohit, Dimitrios Athinaios

Apologies: None

Review of Work Completed:

- Abhishek - Demonstrated working web scraper extracting net-zero years and reduction percentages. Identified JavaScript rendering issues with GSK, AstraZeneca, and M&S; proposed Wayback Machine fallback.

- Vedant - Presented Dockerfile, run_pipeline.sh, run_pipeline.bat, and .gitignore. Docker container building successfully on macOS. Began OWID CO2 data ingestion module.
- Salman - Reddit API module in progress — rate limiting strategy discussed and agreed at 3–5 seconds. Guardian API initial queries returning articles. Shared company list with sectors.
- Morvica - Spark pipeline architecture design started. NetworkX graph construction started; edge attribute schema under discussion.
- Dimitrios - SQLite schema drafted with company_claims and sector_summary tables. Data validation rules defined for net-zero year range (2025–2060) and reduction percentage bounds (10–99%).

Decisions:

- Agreed to use Wayback Machine for JS-rendered pages rather than Selenium to avoid browser dependency.
- Agreed on 2-second minimum scraping delay; 3–5 seconds for APIs.
- Abhishek to refine data quality validation filters for values extracted during scraping.

Action Points: Salman to complete Reddit module by Thursday. Morvica to finalise graph edges and begin Spark integration. Vedant to complete OWID ingestion and begin DuckDB design.

Meeting 3 - Friday 7 March 2026, 10:00–11:30 (Online, Microsoft Teams)

Attendees: Salman Khalid, Abhishek Sharma, Vedant Bhatia, Morvica Purohit, Dimitrios Athinaios

Apologies: None

Review of Work Completed:

- Salman - Reddit module complete (664 posts collected across 8 subreddits). Social signal aggregation table built. All data stored to MongoDB and Parquet.
- Morvica - Spark join and risk score logic implemented. Five Spark SQL queries written and tested. NetworkX graph finalised with company, sector, and certification nodes (21 total). GEXF export tested in Gephi successfully.
- Vedant - OWID CO2 data ingestion complete with UK filter applied. Resolved Windows winutils issue with Hadoop configuration. Began DuckDB analytical layer design.
- Abhishek - Web scraping complete for all 10 companies (52 URLs). Data quality validation filters applied. Automation scripts (run_pipeline.sh, run_pipeline.bat) drafted.
- Dimitrios - SQLite database implemented with typed schema and sector_summary aggregation. Data lineage logger implemented. Data validation pipeline tested across all five sources, confirming record counts and schema consistency.

Task Allocations:

- Vedant - Build DuckDB analytical layer with seven queries including CTE-based reality check. Begin Streamlit dashboard development with Plotly visualisations.

- Abhishek - Finalise automation scripts and test full pipeline execution end-to-end.
- Morvica - Finalise Docker containerisation and README documentation.
- Salman - Write the 4,000-word group report.
- Dimitrios - Conduct end-to-end data quality audit and verify Parquet schema consistency across all pipeline outputs.

Action Points: All code committed by Saturday evening. Full pipeline test run Saturday night. Salman to circulate report draft Sunday for review.

Meeting 4 - Saturday 8 March 2026, 20:00–21:00 (Online, Microsoft Teams)

Attendees: Salman Khalid, Abhishek Sharma, Vedant Bhatia, Morvica Purohit, Dimitrios Athinaios

Apologies: None

Review of Work Completed:

- Vedant - DuckDB queries complete including CTE-based reality check with LAG() window function. All outputs saved as Parquet. Streamlit dashboard functional with 8 tabs, Plotly interactive visualisations, and risk-based colour coding. Dark theme implemented.
- Morvica - Docker containerisation finalised. Full pipeline tested end-to-end in Docker container. README documentation updated.
- Abhishek - Automation scripts verified on macOS and Windows. Requirements configuration finalised. Full end-to-end pipeline test completed successfully.
- Salman - Report first draft complete. Presented key finding: all 10 companies hold certifications (SBTi universal), capping risk scores at 1. Retail sector claims highest reductions (avg 85%).
- Dimitrios - Data quality audit complete. Verified all Parquet files load correctly in both Spark and DuckDB. Confirmed SQLite schema integrity and sector_summary aggregation accuracy. Proofread report draft and provided feedback on technical accuracy.

Decisions:

- Agreed to merge all feature branches to main and verify final commit history shows contributions from all members.
- Salman nominated to submit the final PDF on behalf of the group by 12 March deadline.
- Group reviewed report draft and provided feedback - all comments incorporated.

Action Points: Salman to finalise report and convert to PDF. Vedant to verify teaching team access to GitHub. All members to review final submission before upload.

Appendix B: Source Version Control

The project uses Git for source version control, hosted on a private GitHub repository (https://github.com/vedantbhatiaa/greenwashing_data_extraction). All five team members contributed code through individual commits, demonstrating iterative development across data collection, processing, analytics, infrastructure, and documentation modules.

Note: the current repository was recreated on 11 March 2026 after the original repository required replacement due to an incorrect repository name in the initial commit that could not be removed without rebasing 40+ subsequent commits. All code was transferred from the original private repository; the original development timeline spanning 4–10 March 2026 is corroborated by the meeting minutes in Appendix A and the Trello board task progression in Appendix C. The commit history below shows contributions from all team members:

- **Vedant Bhatia** (vedantbhatiaa) - DuckDB queries, OWID data ingestion, Spark SQL analysis, dashboard implementation, initial repository setup
- **Morvica Purohit** (morvicapurohit) - Distributed processing (Spark pipeline), Docker containerisation, README documentation
- **Abhishek Sharma** (Abhi4091-art) - Web scraping module, automation scripts (run_pipeline), requirements configuration
- **Salman Khalid** (salmankhalid-hub) - Reddit data collection, README updates, report writing and compilation
- **Dimitrios Athinaios** (daththeanalyst) - SQLite relational schema design, data validation and quality assurance, Parquet schema verification, report proofreading

Commits		
main	All users	All time
Commits on Mar 11, 2026		
UPDATE	vedantbhatiaa committed 33 minutes ago	d723a21
changing filename	vedantbhatiaa committed 42 minutes ago	3da6b98
Merge branch 'main' of https://github.com/vedantbhatiaa/greenwashing_data_extraction	vedantbhatiaa committed 44 minutes ago	2a8aba6
UPDATE	vedantbhatiaa committed 44 minutes ago	1f95041
Update README	morvicapurohit authored 46 minutes ago	Verified cbe61ee
Update	morvicapurohit authored 50 minutes ago	Verified ec3f801
Merge branch 'main' of https://github.com/vedantbhatiaa/greenwashing_data_extraction	vedantbhatiaa committed 1 hour ago	5c28788
UPDATE	vedantbhatiaa committed 1 hour ago	8dfbb65
Update README	morvicapurohit authored 1 hour ago	Verified 6ad9261
Added Requirements txt file	daththeanalyst authored 1 hour ago	Verified 96d1e1b
UPDATE	vedantbhatiaa committed 1 hour ago	73e2c67
Data Lineage	vedantbhatiaa committed 1 hour ago	cc4e784
Analysis		92b4f72

Figure 6: GitHub commit history (page 1) - showing commits from vedantbhatiaa, morvicapurohit, Abhi4091-art, and salmankhalid-hub and daththeanalyst on 11 March 2026.

Analysis vedantbhatiaa committed 1 hour ago	92b4f72		
Query analysis vedantbhatiaa committed 1 hour ago	629f2d2		
DuckDB vedantbhatiaa committed 1 hour ago	0d30eb6		
Update morvicapurohit committed 2 hours ago	8069bbe		
Distributed processing morvicapurohit committed 2 hours ago	567323a		
OWID Data vedantbhatiaa committed 2 hours ago	6feb9c8		
Reddit scraping SalmanKhalid-hub committed 2 hours ago	7494209		
Web scraping Abhi4091-art authored 3 hours ago	Verified c5eb6fa		
UPDATE README SalmanKhalid-hub authored 3 hours ago	Verified 9c91f4d		
UPDATE Abhi4091-art authored 3 hours ago	Verified b93d3fb		
UPDATE vedantbhatiaa authored 3 hours ago	Verified 9e94d73		
Adding run pipeline Abhi4091-art authored 3 hours ago	Verified a291224		
Adding Docker file morvicapurohit authored 3 hours ago	Verified 7bd6ed0		
Adding dashboard.py vedantbhatiaa authored 3 hours ago	Verified 8b2d1ad		
Initial commit vedantbhatiaa authored 3 hours ago	Verified fcdd63a		

Figure 7: GitHub commit history (page 2) - showing additional commits including web scraping, distributed processing, Docker setup, dashboard, and initial commit.

Appendix C: Project Management

Task management was coordinated using a Trello board (MSIN0166 Data Engineering Group Project) following a Kanban workflow with five columns: To Do, Doing, Done, Backlog, and In Review. Tasks were created for each pipeline component and moved through the workflow as development progressed. Key completed tasks included: project scoping and TA consultation, Git repository and Docker configuration, data lineage implementation, web scraper development, data source documentation, two data collection pipeline builds, data validation and QA, and data design documentation. Active tasks at time of screenshot included NetworkX graph construction, Spark pipeline development, Pandas fallback implementation, and visualisation generation.

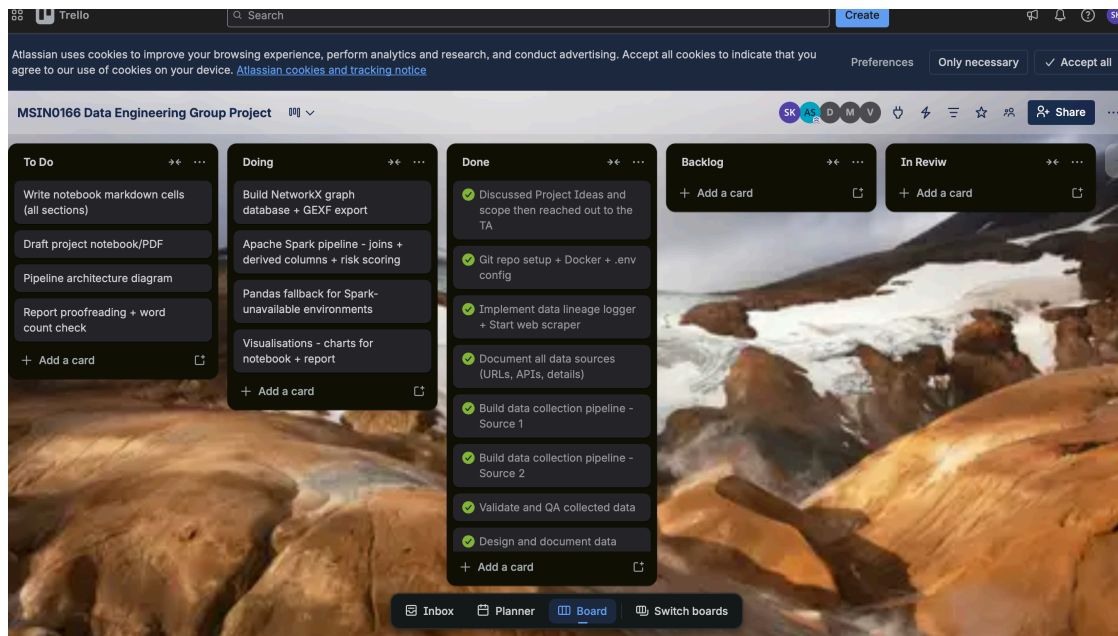


Figure 8: Trello project management board - Kanban workflow tracking pipeline development tasks across To Do, Doing, Done, Backlog, and In Review columns.